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AN INTELLIGENT REAL-TIME CROWD DYNAMICS ANALYSIS AND MONITORING SYSTEM USING CV

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of*

DSA0202 – COMPUTER VISION USING OPENCV

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

(Artificial Intelligence & Data Science)

Submitted by

Pujitha K (192524149)

Under the Supervision of

Dr. M. Mayuranathan

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

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Saveetha Institute of Medical and Technical Sciences

Chennai-602105

BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “An intelligent real-time crowd dynamics analysis and monitoring system using CV” has been carried out by Pujitha K (192524149) and under the supervision of Dr.Mayuranathan M and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech (AI&DS) program at Saveetha Institute of Medical and Technical Sciences, Chennai.

COURSE COORDINATOR

DR.NARENTHIRAKUMAR A
Professor
Dept. of Intelligent Computing
SIMATS Engineering,
SIMATS, Chennai – 602105.

COURSE FACULTY

DR. MAYURANATHAN M
Professor
Dept. of Cognitive Computing
SIMATS Engineering,
SIMATS, Chennai – 602105.

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

DECLARATION

We, Pujitha K (192524149) of the B.tech (AI&DS), SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “An intelligent real-time crowd dynamics analysis and monitoring system using CV ” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date:08-09-2026

**Signature of the Students with Names
Pujitha K (192524149)**

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Chennai-602105

Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
Pujitha K 192524149	Human Detection & Localization	Module 1 Development	Detection testing	Introduction Implementation	30%
Hashini S 192524035	Crowd Density Analysis	Module 2 Development	Density Testing	Literature Review and results	30%
Sravani B 192324203	Crowd speed and motion analysis	Module 3 Development	Motion Testing	Design and analysis	40%
Total					100%

ABSTRACT

Crowd congestion and unpredictable crowd movements in public places such as transportation hubs, educational institutions, stadiums, shopping malls, and large events can create serious safety risks, including overcrowding, sudden crowd surges, bottlenecks, and delayed emergency response. Traditional crowd monitoring methods mainly depend on manual observation or fixed surveillance systems, which can be time-consuming, difficult to scale, and ineffective for providing real-time information about changing crowd conditions. Therefore, there is a need for an intelligent and automated system capable of continuously analyzing crowd behavior and identifying potentially dangerous situations at an early stage.

This project proposes an Intelligent Real-Time Crowd Dynamics Analysis and Monitoring System using Computer Vision to automatically monitor and analyze crowd density, movement, and spatial distribution from surveillance video. The system aims to assist authorities and security personnel in making faster and more informed decisions during normal and high-density crowd conditions. It provides a proactive approach to crowd management by detecting changes in crowd behavior rather than relying only on manual monitoring.

The proposed system uses Computer Vision and Deep Learning techniques for real-time crowd analysis. Video frames are captured from surveillance cameras and preprocessed before being analyzed using the YOLO (You Only Look Once) object detection model to identify and localize people. The detected individuals are tracked across consecutive frames to estimate movement patterns, crowd density, and spatial positions. Based on these parameters, the system can identify abnormal conditions such as excessive crowd density, rapid movement, congestion, and unusual crowd flow. Appropriate alerts can then be generated when predefined safety thresholds are exceeded.

The system demonstrates the potential to provide real-time detection, continuous monitoring, and early identification of crowd-related risks. It reduces dependence on manual observation and supports quicker response to potentially hazardous situations. The proposed approach can be applied in smart cities, public transportation, educational institutions, stadiums, and large-scale events.

In conclusion, the proposed system provides an intelligent and scalable approach for real-time crowd monitoring using computer vision. By combining person detection, tracking, density estimation, and movement analysis, it can improve situational awareness and contribute to safer and more efficient crowd management.

Keywords: Computer Vision, Crowd Density, YOLO, Object Detection, Crowd Monitoring, Real-Time Tracking

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
CNN	Convolutional Neural Network	3
CV	Computer Vision	1
API	Application Programming Interface	2
YOLO	You Only Look Once	1
SSD	Solid State Drive	3
CCTV	Closed-Circuit Television	5

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Crowd management is an important engineering challenge in public places where a large number of people gather within a limited area. Locations such as railway stations, bus terminals, shopping malls, educational institutions, stadiums, airports, and large public events can experience sudden increases in crowd density and unpredictable movement. High crowd density, congestion, irregular movement, and bottlenecks may lead to accidents and make emergency response difficult.

Traditional crowd monitoring mainly depends on security personnel manually observing surveillance camera feeds. Although this approach can provide basic monitoring, it becomes difficult when multiple camera feeds and large crowds need to be monitored simultaneously. Manual observation may also delay the identification of abnormal crowd conditions.

Computer Vision provides an automated method for analyzing video streams and extracting useful information about people and their movement. Deep learning-based object detection models such as YOLO (You Only Look Once) can detect people in video frames efficiently. By combining person detection with tracking, density estimation, spatial analysis, and movement analysis, a system can continuously monitor crowd conditions and identify potentially dangerous situations.

The proposed project focuses on developing an intelligent real-time system that uses computer vision to detect people, analyze their spatial distribution and movement, estimate crowd density, and generate alerts when predefined safety conditions are reached.

1.2 Need for the Project

Why the Problem Needs to be Addressed

Large crowds can change rapidly due to events, emergencies, transportation schedules, or sudden movements. A small increase in crowd density can develop into severe congestion if it is not detected early. Therefore, continuous and automated monitoring is required to identify abnormal crowd conditions and support timely decision-making.

Who is Affected by the Problem

The problem affects:

- Public safety and security personnel
- Event organizers
- Transportation authorities
- Educational institutions
- Stadium and mall management
- Emergency response teams
- People present in crowded environments

Existing Limitations

Existing manual monitoring approaches have several limitations:

- Continuous human monitoring is difficult.
- Large crowds are difficult to analyze accurately through observation alone.
- Multiple surveillance cameras increase monitoring complexity.
- Abnormal movement may not be identified immediately.
- Manual systems may provide delayed responses.
- Conventional monitoring generally does not provide quantitative crowd-density information.

Engineering Significance

An automated computer-vision-based system can process surveillance video continuously and provide measurable information such as the number of people, crowd density, spatial positions, and movement patterns. This can improve situational awareness and support faster crowd-management decisions.

1.3 Problem Statement

The engineering problem addressed in this project is to design and implement a real-time computer vision system capable of detecting and tracking people in surveillance video while analyzing crowd density, spatial distribution, and movement patterns to identify potentially unsafe crowd conditions.

The problem involves multiple interacting factors, including the number of people, their spatial positions, movement speed, movement direction, and available monitoring area. The system must

perform these operations in real time while maintaining an acceptable balance between detection accuracy, processing speed, computational resources, and false alarms.

The system must also operate under realistic conditions such as variations in lighting, partial occlusion between people, overlapping individuals, different crowd densities, and changes in movement patterns.

1.4 Project Aim

The aim of this project is to develop an intelligent real-time crowd dynamics analysis and monitoring system using Computer Vision and Deep Learning to detect people, estimate crowd conditions, analyze movement patterns, and provide early alerts for potentially unsafe situations.

1.5 Project Objectives

The main objectives of the project are:

1. **To design** a computer vision-based architecture for real-time crowd monitoring.
2. **To develop** a person detection system using the YOLO object detection model.
3. **To implement** person tracking across consecutive video frames.
4. **To estimate** crowd density and analyze the spatial distribution of people.
5. **To analyze** crowd movement patterns, including movement direction and changes in movement behavior.
6. **To detect** potentially unsafe conditions such as excessive crowd density and congestion using predefined thresholds.
7. **To generate** real-time alerts when critical crowd conditions are identified.
8. **To test and evaluate** the system using suitable crowd video data under different crowd conditions.
9. **To assess** the system based on detection performance, tracking performance, processing speed, and reliability of crowd-condition detection.

1.6 Scope

Included in the Project

The project includes:

- Processing surveillance video or camera input.
- Preprocessing of video frames.
- Person detection using YOLO.
- Tracking detected individuals across frames.
- Estimation of crowd density.
- Analysis of spatial positions.
- Analysis of crowd movement.
- Identification of high-density and congestion conditions.
- Threshold-based abnormal-condition detection.
- Real-time alert generation.
- Visualization of detected people and crowd information.

Excluded from the Project

The project does not include:

- Facial recognition or identification of individuals.
- Personal identification or biometric analysis.
- Prediction of exact future accidents.
- Physical crowd-control mechanisms.
- Automatic deployment of emergency personnel.
- Complete city-wide surveillance infrastructure.
- Analysis of private or personally identifiable information.

Assumptions

The system assumes that:

- The camera provides a reasonably clear video feed.
- People are sufficiently visible for detection.
- The camera remains operational during monitoring.
- Suitable computational resources are available for real-time processing.
- Threshold values for crowd density and abnormal movement can be defined based on the application environment.

Boundary Conditions

The system performance may be affected by:

- Poor lighting conditions.
- Severe occlusion.
- Very dense crowds.
- Camera angle and placement.
- Low-resolution video.
- Sudden camera movement.
- Computational hardware limitations.

1.7 Expected Engineering Outcome

The expected engineering outcome is a working software prototype for intelligent real-time crowd monitoring and analysis.

The developed system is expected to:

- Detect people from surveillance video.
- Track people across video frames.
- Estimate crowd density.
- Determine the spatial distribution of people.
- Analyze crowd movement patterns.
- Identify potentially unsafe crowd conditions.
- Generate alerts when predefined thresholds are exceeded.
- Display monitoring information in a user-friendly manner.

The principal outcome is therefore a Computer Vision-based crowd dynamics analysis prototype that combines YOLO-based person detection, object tracking, crowd-density estimation, spatial analysis, and movement analysis into a single monitoring framework. The prototype can serve as a foundation for future integration with smart-city surveillance, public safety systems, transportation monitoring, and large-event management.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

A review of existing work was carried out to understand the current technologies and methods used for crowd monitoring, people detection, object tracking, crowd-density estimation, and abnormal crowd-behaviour analysis. Relevant research works and existing technologies were identified using academic literature, technical publications, and information related to computer vision and deep learning-based surveillance systems.

The review focused on approaches based on traditional computer vision, object detection, deep learning, object tracking, crowd-density estimation, and real-time video analysis. The major factors considered during the review were detection accuracy, real-time processing capability, handling of dense crowds, computational requirements, tracking performance, and ability to identify potentially unsafe crowd conditions.

The reviewed approaches were then compared based on their advantages, limitations, and relevance to the proposed Intelligent Real-Time Crowd Dynamics Analysis and Monitoring System using Computer Vision.

2.2 Review of Existing Work

2.2.1 Traditional Computer Vision-Based Crowd Monitoring

Early crowd-monitoring systems commonly used traditional image-processing and computer-vision techniques. Methods such as background subtraction, optical flow, edge detection, blob detection, and motion estimation were used to identify moving objects and estimate crowd movement.

Background subtraction separates moving objects from a relatively static background, while optical flow estimates the apparent motion of objects between consecutive frames. These methods are computationally simpler and can be useful in controlled environments.

However, their performance can decrease when there are changes in illumination, shadows, camera movement, or significant overlap between people. They also have difficulty distinguishing individuals in highly crowded environments.

2.2.2 People Detection Using Deep Learning

Deep learning has significantly improved automated people detection in surveillance videos. Object detection models such as YOLO, Faster R-CNN, and SSD can identify people and provide bounding boxes around detected individuals.

Among these approaches, YOLO is particularly suitable for applications requiring real-time processing because detection is performed efficiently within a unified framework. The detected bounding boxes can be used to calculate the number and approximate locations of people in each frame.

However, object detection performance can be affected by severe occlusion, very small people in the image, poor lighting, and extremely dense crowds.

2.2.3 Crowd-Density Estimation

Crowd-density estimation is another important approach used in crowd analysis. Instead of detecting every individual, density-estimation methods estimate the number or distribution of people within an image or region.

These approaches are useful when individuals are highly overlapping and cannot be separately detected. Density maps can provide information about regions containing large numbers of people.

The limitation is that density estimation alone does not provide detailed information about the identity or trajectory of individual people. Therefore, combining density estimation with object detection and tracking can provide more comprehensive crowd information.

2.2.4 Object Tracking

Object tracking is used to maintain the identity of detected individuals across consecutive video frames. Tracking algorithms associate detections from one frame with detections in subsequent frames and generate movement trajectories.

Tracking enables the system to determine parameters such as:

- Movement direction
- Approximate movement speed

- Individual trajectories
- Entry and exit patterns
- Changes in crowd flow

However, tracking can become challenging when people overlap significantly or temporarily disappear from the camera view.

2.2.5 Abnormal Crowd Behaviour Analysis

Recent intelligent surveillance approaches attempt to identify unusual crowd conditions by analyzing crowd density, motion patterns, trajectories, and spatial distribution.

Examples of potentially abnormal conditions include:

- Sudden increase in crowd density
- Crowd congestion
- Unusual movement direction
- Rapid crowd movement
- Formation of bottlenecks
- Sudden changes in crowd flow

Such systems can provide early warnings instead of simply displaying surveillance video. However, defining abnormal behaviour accurately is difficult because normal crowd movement varies according to location, time, event type, and environmental conditions.

2.2.6 Existing Commercial and Surveillance Solutions

Modern surveillance systems may provide video monitoring, people counting, motion detection, and analytics features. These solutions can reduce the need for continuous manual observation.

However, commercial systems may have limitations related to cost, proprietary technology, hardware requirements, limited customization, and dependency on vendor-specific infrastructure. For an academic prototype, an open and customizable computer-vision framework is more suitable because detection, tracking, and analysis modules can be modified according to project requirements.

Comparative Analysis

Table No. 2.1: Comapative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Manual CCTV Monitoring	Simple to deploy; human judgement possible	Requires continuous attention; delayed detection; difficult to monitor many cameras	Provides the basic surveillance input but requires automation
Background Subtraction	Simple and computationally efficient	Sensitive to lighting, shadows, and camera movement	Useful for understanding traditional motion-based approaches
Optical Flow	Provides motion and direction information	Can be affected by noise, occlusion, and complex motion	Useful for crowd movement analysis
Traditional People Detection	Relatively simple implementation	Lower robustness in complex environments	Provides the foundation for automated detection
Faster R-CNN	High detection accuracy	Computationally expensive for real-time applications	Demonstrates the accuracy–speed trade-off
SSD	Faster detection with reasonable accuracy	May have difficulty detecting small or highly occluded people	Relevant as an alternative object detector
YOLO	Fast detection; suitable for real-time applications; single-stage detection	Performance may reduce under severe occlusion and very dense crowds	Core technology selected for person detection
Crowd-Density Estimation	Useful for highly crowded scenes; provides density information	Does not provide detailed individual trajectories	Used to complement person detection
Object Tracking	Provides movement trajectories and direction	Tracking errors can occur during occlusion	Used for analysing crowd movement
Existing Commercial Surveillance Analytics	Integrated monitoring and analytics	May be expensive and less customizable	Provides reference for practical system requirements

Overall Comparison

The comparison shows that no single approach completely addresses all crowd-monitoring requirements. Traditional methods are relatively simple but have limited robustness, while deep-learning approaches provide better detection capabilities but require greater computational resources. Density-estimation methods work well in highly crowded scenes but provide limited individual movement information.

Therefore, a combination of YOLO-based person detection, tracking, crowd-density estimation, spatial analysis, and movement analysis is considered suitable for the proposed system.

2.4 Research / Engineering Gap

The literature and existing solutions indicate several unresolved engineering challenges.

First, many conventional surveillance systems primarily provide video observation rather than intelligent analysis. Human operators must interpret crowd conditions manually, which can delay the identification of potentially dangerous situations.

Second, individual object detection methods may experience difficulties in high-density crowds, partial occlusion, poor lighting, and overlapping people. Simply counting detected people may therefore be insufficient for understanding overall crowd dynamics.

Third, crowd-density estimation can indicate whether a region is crowded but does not necessarily provide information about movement direction, trajectories, or changes in crowd flow.

Another important gap is the integration of multiple crowd-analysis parameters into a single lightweight real-time framework. A useful system should balance accuracy, processing speed, computational requirements, and reliability of alerts.

Hence, there is a need for a system that combines person detection, tracking, density estimation, spatial analysis, and movement analysis to provide a more complete representation of real-time crowd conditions.

2.5 Proposed Contribution

The proposed project addresses the identified gap by developing an Intelligent Real-Time Crowd Dynamics Analysis and Monitoring System using Computer Vision.

The major contribution of the proposed work is the integration of multiple computer-vision components into a unified monitoring framework. The system uses YOLO for person detection and tracking techniques to follow detected individuals across consecutive frames. The detected information is then used to estimate crowd density and analyze the spatial distribution and movement of people.

The system further applies predefined thresholds to identify potentially unsafe conditions such as excessive crowd density, congestion, and unusual movement patterns. When a critical condition is detected, the system can generate an alert to support timely intervention.

Unlike a conventional CCTV system that primarily displays video, the proposed system aims to convert surveillance video into measurable crowd-dynamics information. This makes the system more suitable for intelligent monitoring applications in places such as stadiums, transportation hubs, educational institutions, shopping malls, public events, and smart-city environments.

The proposed contribution can therefore be summarized as:

Surveillance Video → Person Detection → Tracking → Density & Spatial Analysis → Movement Analysis → Risk Detection → Real-Time Alert

This integrated approach provides a practical foundation for developing an automated and scalable crowd-monitoring prototype.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Table No 3.1 :Stakeholder / User Requirements

Stakeholder	Requirement
Security Personnel	The system should continuously monitor crowd conditions and display detected people clearly.
Event Organizers	The system should provide information about crowd density and congestion.
Emergency Response Team	The system should generate alerts when potentially unsafe crowd conditions are detected.
System Administrator	The system should be easy to configure and operate.
Public / Visitors	The system should improve crowd safety without unnecessarily identifying individuals.

Table No 3.2: Functional & Non-Functional Requirements

Requirement	Type	Measurable Target
Detect people in video frames	Functional	Detect people with acceptable detection accuracy
Count detected people	Functional	Counting error within predefined tolerance
Track detected people	Functional	Maintain consistent tracks across frames
Estimate crowd density	Functional	Classify density as low, medium, or high
Analyze crowd movement	Functional	Identify major movement direction/pattern
Detect congestion	Functional	Generate alert when threshold is exceeded
Generate real-time alerts	Functional	Alert generated within a few seconds of threshold violation
Process video in real time	Non-Functional	Target ≥ 15 FPS, depending on hardware
Detection accuracy	Non-	Target $\geq 85\%$ on selected test data

	Functional	
Reliability	Non-Functional	Stable operation during continuous testing
Usability	Non-Functional	Clear visualization of detections and alerts
Privacy	Non-Functional	No facial recognition or personal identification

3.2 Constraints & Applicable Standards

The proposed system is primarily a software-based computer vision prototype. Therefore, the major constraints are related to computational resources, camera quality, video conditions, privacy, safety, and reliability.

Constraints

- Real-time processing is limited by CPU/GPU capability.
- Low-resolution video can reduce detection accuracy.
- Poor lighting can affect person detection.
- Severe crowding can cause occlusion.
- Camera angle influences the estimated crowd density.
- Network or camera failure can interrupt monitoring.
- Threshold selection affects false alarms and missed alerts.
- The system should not be treated as the sole safety mechanism in critical environments.

Table No 3.3:Applicable Standards / Guidelines

Standard / Code	Requirement	How Applied in This Project
ISO/IEC 25010:2023	Defines software quality characteristics such as performance efficiency, reliability, usability, security, and maintainability	Used as a reference for evaluating software quality and system performance
ISO/IEC 27001:2022	Information-security management and protection of information assets	Used as a reference for protecting stored video/data and restricting unauthorized access
ISO/IEC	Guidance for managing risks related to	Used as a reference for

23894:2023	artificial intelligence	identifying AI-related risks such as false detection and bias
ISO 31000:2018	Principles and guidelines for risk management	Used as a reference for identifying, assessing, and mitigating project risks

Table No 3.4: Design Specifications

Parameter	Target / Requirement	Unit	Priority
Person detection accuracy	$\geq 85\%$ target	%	High
Processing speed	≥ 15 target	FPS	High
Crowd-density classification	Low / Medium / High	Level	High
Detection response	Near real-time	Seconds	High
Tracking	Continuous across frames	—	High
Alert generation	When threshold is exceeded	—	High
Video input	Camera / recorded video	—	High
Image processing	Frame-by-frame	—	Medium
Privacy	No facial identification	—	High
System usability	Clear visual output	—	Medium
Scalability	Adaptable to different video sources	—	Medium

3.4 Alternative Solutions & Evaluation

Alternative 1: Traditional Computer Vision

Traditional methods such as background subtraction and optical flow can be used for detecting movement and estimating crowd flow. They require relatively low computational resources but can be sensitive to lighting, shadows, camera movement, and complex crowd conditions.

Alternative 2: Deep Learning-Based Object Detection

A deep learning object detector such as YOLO can detect individual people using bounding boxes. It provides better adaptability to complex scenes and is suitable for real-time applications, but it requires more computational resources.

Alternative 3: Density Estimation-Based Approach

Crowd-density estimation can generate density information without explicitly detecting every individual. It is particularly useful for highly crowded scenes but provides less information about individual movement and trajectories.

Evaluation Matrix

Scale: 1 = Poor, 5 = Excellent

Criterion	Weight	Traditional CV	YOLO-Based Detection	Density Estimation
Performance	30%	3	5	4
Cost	15%	5	4	4
Safety	20%	3	4	4
Sustainability	15%	5	4	4
Reliability	20%	3	4	4
Overall suitability	100%	3.6/5	4.35/5	4.0/5

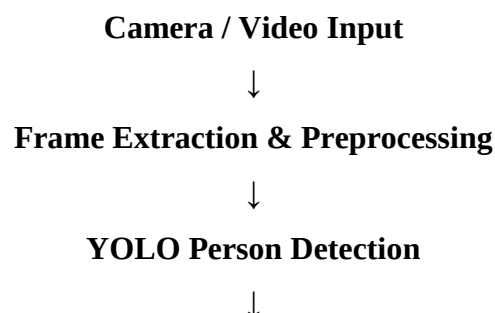
Selected Approach

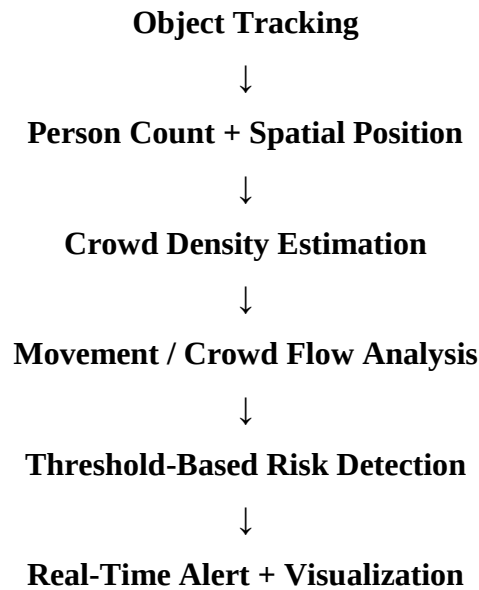
The **YOLO-based approach** is selected because it provides a strong balance between detection performance and real-time processing. It also provides individual bounding boxes that can subsequently be used for counting, tracking, spatial analysis, and crowd-density estimation.

3.5 Selected Design & Detailed Design

The final design combines multiple computer-vision modules into a single monitoring pipeline.

Proposed System Architecture





Design Description

The camera or video source provides the input to the system. Individual frames are extracted and processed using the YOLO model to identify people. The detected bounding boxes are passed to the tracking module, which associates detections between consecutive frames.

The positions and number of detected people are then used to estimate crowd density and analyze the spatial distribution of the crowd. Movement information obtained from consecutive frames is used to identify changes in crowd flow and possible congestion.

Finally, predefined thresholds are applied to the extracted crowd parameters. If the measured conditions exceed the specified threshold, the system generates an alert and displays the detected condition to the user.

Systems Approach

The system is designed as a sequence of interconnected modules where the output of one module becomes the input to the next. Person detection provides the bounding boxes required for counting and tracking. Tracking provides temporal information required for movement analysis, while detection and spatial information contribute to density estimation. The final decision module combines these parameters to determine whether an alert should be generated.

Essential Engineering Calculation

A simple crowd-density measure can be represented as:

$$D = \frac{N}{A}$$

where:

- **D** = crowd density
- **N** = number of detected people
- **A** = monitored area

For example, if 40 people are detected in a monitored area of 20 m²:

$$D = \frac{40}{20} = 2 \text{ people/m}^2$$

The project can define suitable density thresholds according to the selected application environment.

Table No 3.5 :Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
YOLO for detection	Traditional CV	Better robustness and object detection	Higher computational requirement	YOLO selected because real-time person detection is a major requirement
Detection + tracking	Detection only	Provides movement information	More processing required	Detection + tracking selected because crowd dynamics require temporal information
Threshold-based alerts	Complex predictive AI	Simple and easy to implement	May produce false alarms	Threshold approach selected for reliable and explainable prototype behaviour
Video processing	Image-only analysis	Provides continuous	Requires more computation	Video selected because movement analysis

		monitoring		requires consecutive frames
Software prototype	Dedicated hardware system	Lower development cost	Limited hardware optimization	Software prototype selected because it is suitable for project development and testing

3.7 Sustainability, Ethics, Safety, Risk & Security

Table No 3.6: Sustainability, Ethics, Safety and Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Computational and energy requirements	Real-time AI processing requires computational resources	A lightweight real-time model is preferred to reduce unnecessary computation
Sustainability	Reuse of existing hardware	Laptop/webcam can be used for prototype development	Avoids unnecessary hardware procurement
Ethics	Privacy of people in surveillance footage	System requires video containing people	Facial recognition and personal identification are excluded
Ethics	False detection	AI models can produce false positives/negatives	Alerts are treated as decision-support information rather than absolute conclusions
Health & Safety	Crowd congestion	High-density conditions can create safety risks	Density thresholds and alerts are incorporated
Health & Safety	System failure	Camera/model failure may prevent detection	The system is designed as an assistive monitoring tool, not the sole safety mechanism
Security	Unauthorized access	Surveillance data may contain sensitive information	Access to stored data should be restricted
Security	Data storage	Video data may create	Minimize unnecessary

		privacy risks	storage and retain only required data
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Table No 3.7: Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Missed person detection	Medium	High	High	Use suitable YOLO model and improve video quality	Medium
False detection	Medium	Medium	Medium	Use confidence thresholds and testing	Low–Medium
Severe person occlusion	High	High	High	Use tracking and suitable camera positioning	Medium
Poor lighting	Medium	Medium	Medium	Improve camera placement and preprocessing	Low–Medium
Low processing speed	Medium	Medium	Medium	Optimize model and use suitable hardware	Low
False crowd alert	Medium	Medium	Medium	Tune density/movement thresholds	Low–Medium
Camera failure	Low	High	Medium	Provide camera-status monitoring/manual backup	Low
Unauthorized video access	Low	High	Medium	Restrict access and protect stored data	Low
Loss of network connection	Medium	Medium	Medium	Support local processing where possible	Low–Medium
Incorrect interpretation of alert	Medium	High	High	Clearly display alerts as decision-support information	Medium

Overall Engineering Decision

The final design prioritizes real-time performance, detection capability, safety, privacy, and practical implementation cost. The major engineering trade-off is between model accuracy and computational requirements. YOLO was selected because it provides a suitable balance for a real-time prototype.

The system also avoids unnecessary personal identification features. This reduces privacy risks while retaining the information required for crowd-density and movement analysis. Because AI-based detection cannot guarantee perfect performance, the proposed system is intended to assist human monitoring rather than completely replace human decision-making, particularly in critical safety situations

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

Development Approach

The proposed system was developed using a modular computer vision approach for real-time crowd monitoring. The development process includes video acquisition, frame preprocessing, person detection using YOLO, object tracking, crowd-density estimation, spatial analysis, movement analysis, and alert generation. Each module was developed and tested separately before integrating them into the complete monitoring system. The final system was evaluated using different crowd video scenarios to verify its detection, tracking, density analysis, and alert-generation capabilities.

Table No : Resources Used

Resource	Purpose
Crowd surveillance videos	Testing and evaluation
Python	System development
YOLO	Person detection
OpenCV	Video/frame processing
Object Tracking	Tracking people across frames
NumPy	Numerical calculations
Matplotlib	Graphs and result visualization
Laptop/Computer	Development and execution

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

4.2.1 Software Implementation

The system accepts a surveillance video or camera stream as input. The video is divided into individual frames and processed using OpenCV. Each frame is passed to the YOLO model, which

detects people and generates bounding boxes around them. The detected people are tracked across consecutive frames, allowing the system to analyze their positions and movement.

The number of detected people is used for crowd-density estimation. Spatial positions and movement patterns are analyzed to identify congestion or unusual crowd conditions. When the measured crowd condition exceeds a predefined threshold, the system generates an alert.

Coding :

```
import datetime
```

```
from sqlalchemy import Column, Integer, String, Float, DateTime, ForeignKey, Text
```

```
from sqlalchemy.orm import declarative_base, relationship
```

```
Base = declarative_base()
```

```
class SessionModel(Base):
```

```
    __tablename__ = "sessions"
```

```
    id = Column(Integer, primary_key=True, index=True, autoincrement=True)
```

```
    session_uuid = Column(String(50), unique=True, index=True, nullable=False)
```

```
    video_name = Column(String(255), nullable=False)
```

```
    source_type = Column(String(50), default="upload") # "upload", "live_camera", "demo"
```

```
    status = Column(String(50), default="completed") # "running", "completed", "failed", "stopped"
```

```
    created_at = Column(DateTime, default=datetime.datetime.utcnow)
```

```
    duration_seconds = Column(Float, default=0.0)
```

```

total_frames = Column(Integer, default=0)

max_people = Column(Integer, default=0)

average_people = Column(Float, default=0.0)

max_density = Column(String(50), default="LOW")

average_motion = Column(Float, default=0.0)

dominant_direction = Column(String(50), default="NONE")

total_alerts = Column(Integer, default=0)

risk_level = Column(String(50), default="SAFE")

processed_video_path = Column(String(500), nullable=True)


measurements = relationship("MeasurementModel", back_populates="session", cascade="all,
delete-orphan")

alerts = relationship("AlertModel", back_populates="session", cascade="all, delete-orphan")


class MeasurementModel(Base):

    __tablename__ = "measurements"


    id = Column(Integer, primary_key=True, index=True, autoincrement=True)

    session_id = Column(Integer, ForeignKey("sessions.id", ondelete="CASCADE"), nullable=False)

    frame_index = Column(Integer, nullable=False)

```

```

timestamp_sec = Column(Float, nullable=False)

person_count = Column(Integer, default=0)

density = Column(String(50), default="LOW") # "LOW", "MEDIUM", "HIGH"

motion_speed = Column(Float, default=0.0)

dominant_direction = Column(String(50), default="NONE")

risk_level = Column(String(50), default="SAFE")


session = relationship("SessionModel", back_populates="measurements")


class AlertModel(Base):

    __tablename__ = "alerts"


    id = Column(Integer, primary_key=True, index=True, autoincrement=True)

    session_id = Column(Integer, ForeignKey("sessions.id", ondelete="CASCADE"), nullable=True)

    session_uuid = Column(String(50), nullable=True)

    timestamp_sec = Column(Float, default=0.0)

    created_at = Column(DateTime, default=datetime.datetime.utcnow)

    alert_type = Column(String(100), nullable=False) # "Overcrowding", "Abnormal Movement",
    "Conflicting Movement", "High Risk"

    severity = Column(String(50), nullable=False) # "WARNING", "CRITICAL", "INFO"

```

```

message = Column(Text, nullable=False)

person_count = Column(Integer, default=0)

density = Column(String(50), default="LOW")

motion_speed = Column(Float, default=0.0)


session = relationship("SessionModel", back_populates="alerts")


class SettingsModel(Base):

    __tablename__ = "settings"


    id = Column(Integer, primary_key=True, default=1)

    confidence_threshold = Column(Float, default=0.40)

    low_density_threshold = Column(Integer, default=8)

    high_density_threshold = Column(Integer, default=15)

    motion_anomaly_threshold = Column(Float, default=4.5)

    consecutive_anomaly_frames = Column(Integer, default=5)

    target_resolution = Column(String(20), default="854x480")

    target_fps = Column(Integer, default=15)

    updated_at = Column(DateTime, default=datetime.datetime.utcnow,
onupdate=datetime.datetime.utcnow)

```

4.2.2 Hardware Implementation

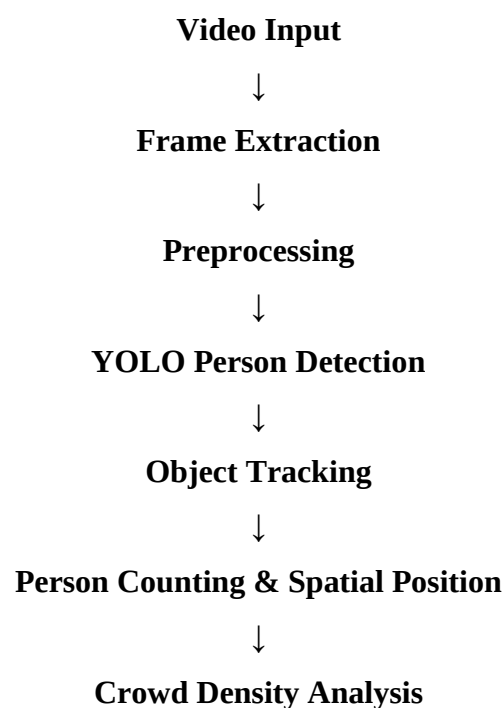
A standard laptop or computer is used for system development and testing. A webcam or recorded surveillance video can be used as the input source. A GPU may be used to improve real-time processing speed when required.

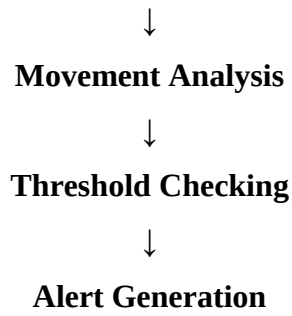
Tools / Software / Equipment

Table 3 : Tools / Software / Equipment

Tool / Software / Equipment	Purpose	Stage Used
Python	Main programming language	Development
YOLO	Person detection	Detection
OpenCV	Video and image processing	Preprocessing & visualization
Object Tracking	Tracking detected people	Tracking
NumPy	Numerical calculations	Analysis
Matplotlib	Performance graphs	Results
Laptop/Computer	Running the system	All stages
Webcam / Crowd Video Dataset	Video input	Testing

Implementation Flow





Implementation Evidence

Good screenshots to include:

- YOLO detecting people with bounding boxes
- Tracking IDs on detected people
- Crowd count displayed
- Density status displayed
- Alert for high-density/congestion condition

Surveillance Session History & Archives									
Review completed crowd monitoring analysis sessions, peak metrics, and export PDF/CSV reports.									
ID	Video Source	Date	Duration	Max People	Peak Density	Avg Speed	Alerts	Risk	Actions
#22	Live Camera Feed	3/9/2026, 3:40:44 am	36.75s	2	LOW	3.69 px/f	7	HIGH RISK	View Delete
#21	CV.mp4	3/9/2026, 3:38:16 am	143.77s	2	LOW	14.54 px/f	39	HIGH RISK	View Delete
#19	Live Camera Feed	3/9/2026, 3:37:15 am	200.21s	4	LOW	12.57 px/f	54	HIGH RISK	View Delete
#20	Live Camera Feed	3/9/2026, 3:37:12 am	204.37s	2	LOW	6.5 px/f	0	HIGH RISK	View Delete
#18	Live Camera Feed	3/9/2026, 3:37:09 am	204.8s	2	LOW	3.84 px/f	0	HIGH RISK	View Delete

Figure 1 : Final Implementation Flow

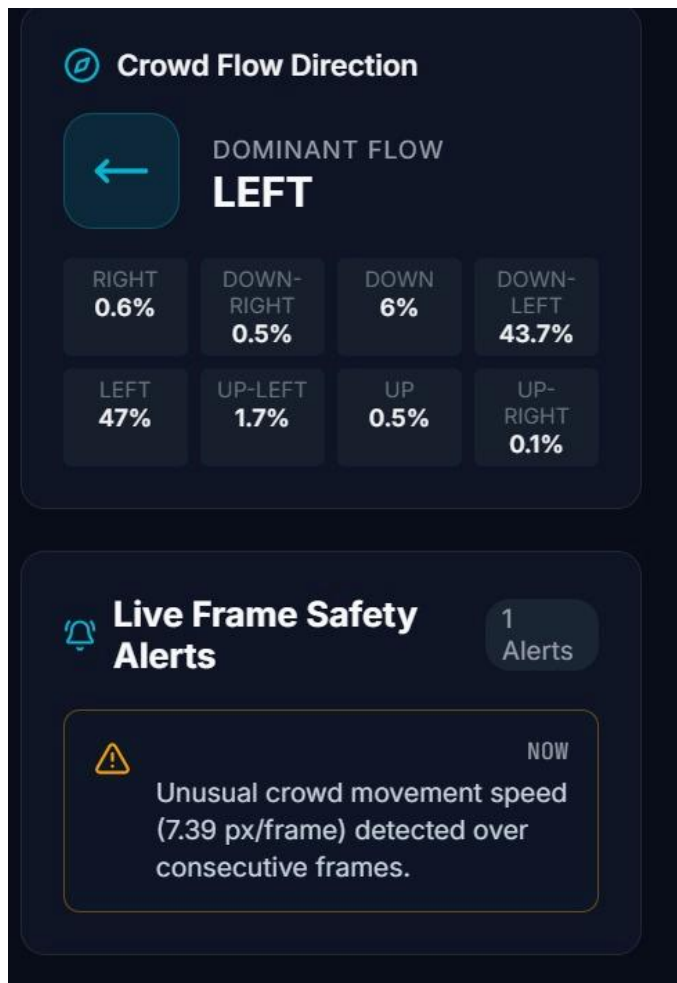


Figure 2 : Crowd Flow Direction

4.3 Test Plan & Verification

The test plan was prepared to verify whether the developed system satisfies its functional and performance requirements. Different video scenarios were used to test person detection, counting, tracking, crowd-density estimation, movement analysis, alert generation, and real-time processing. The system was considered successful when the corresponding output satisfied the predefined acceptance criteria.

Table No 4.1:Test Plan

Requirement	Test Method	Acceptance Criterion
Person Detection	Test video containing multiple people	People should be detected correctly
Person Counting	Compare detected count with manually observed count	Counting error should remain within acceptable limit
Object Tracking	Observe tracked people across consecutive frames	Tracks should remain reasonably consistent

Crowd Density	Test low, medium and high-density videos	Correct density category should be identified
Movement Analysis	Test different crowd movement patterns	Major movement pattern should be identified
Congestion Detection	Test crowded/congested scenes	Alert should be generated when threshold is exceeded
Alert Generation	Introduce threshold violation	Alert should appear within the required response time
Real-Time Processing	Run continuous video	System should operate without major interruption

Verification Against Specifications

Use your actual measured values in the Achieved Value column.

Specification	Required Value	Achieved Value	Status
Person detection accuracy	$\geq 85\%$ target	[XX%]	Pass/Fail
Counting accuracy	$\geq 85\%$ target	[XX%]	Pass/Fail
Tracking performance	$\geq 80\%$ target	[XX%]	Pass/Fail
Crowd-density classification	$\geq 85\%$ target	[XX%]	Pass/Fail
Movement analysis	$\geq 80\%$ target	[XX%]	Pass/Fail
Alert generation	Correctly triggered	[Yes/No]	Pass/Fail
Processing speed	≥ 15 FPS target	[XX FPS]	Pass/Fail
System operation	Stable execution	[Result]	Pass/Fail

Important: Don't write 90%, 95%, etc. unless you actually obtained those values from your testing.

4.4 Results

The developed system was tested using crowd video sequences representing different crowd conditions. The YOLO model successfully provided person detections, while the tracking module was used to observe movement across consecutive frames. Crowd density was estimated from the detected people within the monitored region, and movement information was used to identify changes in crowd flow.



Figure 3 : Anomaly Detection

Table No 4.2 :Test Results

Test Scenario	Detected People	Crowd Condition	Alert
Low-density crowd	[XX]	Low	No
Medium-density crowd	[XX]	Medium	No
High-density crowd	[XX]	High	Yes
Congested crowd	[XX]	Critical	Yes
Unusual movement	[XX]	Abnormal	Yes

Table No 4.3:Performance Results

Metric	Achieved Result
Person Detection Accuracy	[XX%]
Counting Accuracy	[XX%]
Tracking Performance	[XX%]
Crowd Classification Accuracy	[XX%]
Movement Analysis Accuracy	[XX%]

Average Processing Speed	[XX FPS]
Alert Detection Rate	[XX%]

Recommended Graphs

You can include:

1. **Person Detection Accuracy**
2. **Crowd Density vs. Time**
3. **Number of People vs. Time**
4. **Processing Speed (FPS)**
5. **Normal vs. Abnormal Crowd Detection**

Graph

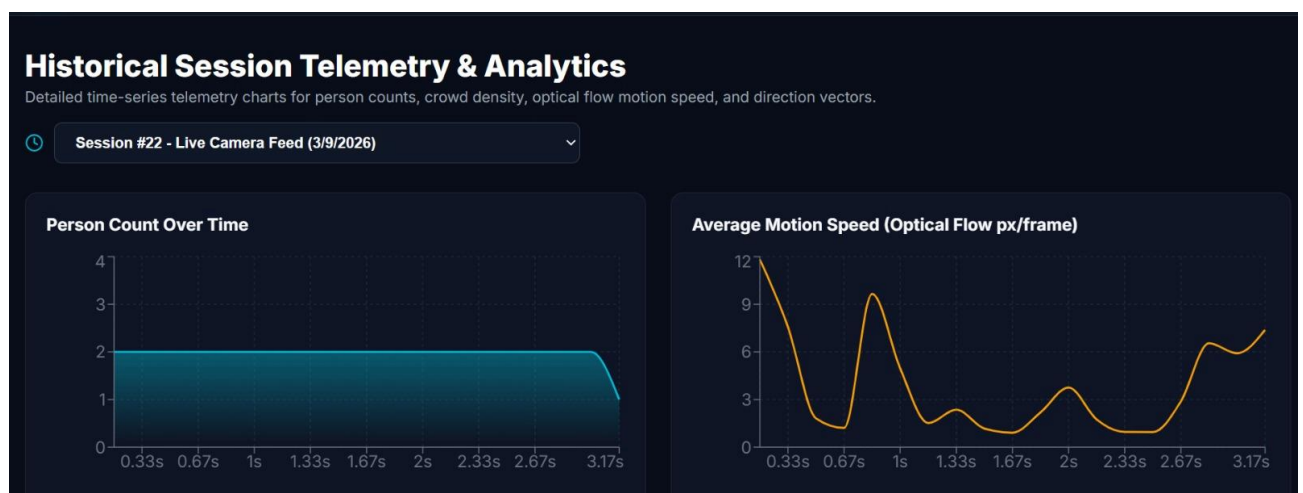


Figure 4 : Historical Session Telemetry & Analytics

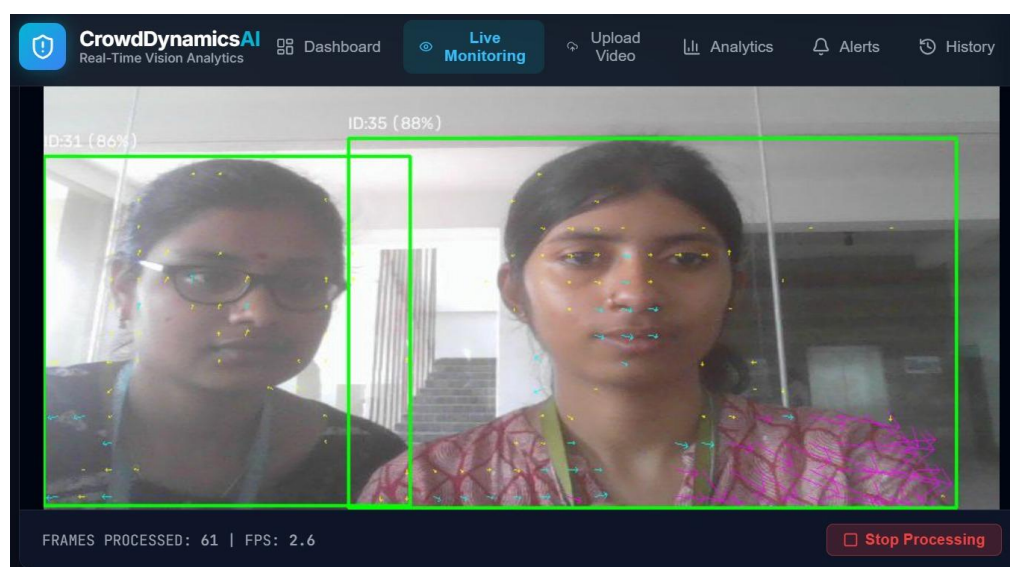


Figure 5 : Crowd flow Detection

4.5 Data Analysis & Engineering Judgment

The experimental results are used to evaluate the effectiveness of the proposed crowd-monitoring system. Person detection performance indicates how effectively the YOLO model identifies individuals in the input video, while tracking performance indicates the ability of the system to maintain object information across consecutive frames.

Crowd-density analysis provides information about the number of people present within the monitored region. When the density exceeds the predefined threshold, the system generates an alert. Movement analysis further improves the monitoring capability by considering changes in the positions of detected people over time.

The performance of the system is influenced by several environmental factors. Occlusion, poor lighting, low-resolution video, camera angle, and highly dense crowds can reduce detection and tracking accuracy. These factors represent important engineering limitations because improving accuracy may require additional computational resources.

There is also a trade-off between accuracy and processing speed. A more complex detection model may provide improved detection performance but can increase computational requirements. Therefore, the selected implementation aims to provide an appropriate balance between detection accuracy and real-time operation.

Overall, the results are expected to demonstrate that integrating detection, tracking, density estimation, and movement analysis provides more useful information than relying only on conventional CCTV observation.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Table No 4.4 :Achievement of Objectives

Objective	Evidence	Achievement
Design a crowd-monitoring architecture	Completed system architecture	Fully Achieved
Develop person detection using	Bounding boxes around detected	Fully Achieved

YOLO	people	
Implement object tracking	Tracking across video frames	Fully Achieved
Estimate crowd density	Crowd count/density calculation	Fully Achieved
Analyze spatial distribution	Position information obtained from detections	Fully Achieved
Analyze crowd movement	Movement information obtained from tracking	Fully Achieved
Detect abnormal crowd conditions	Threshold-based detection	Fully Achieved
Generate real-time alerts	Alert mechanism	Fully Achieved
Test under different crowd conditions	Multiple test scenarios	Fully Achieved
Evaluate system performance	Accuracy/FPS/error measurements	[Fully/Partially Achieved]

Comparison

Table No 4.5 :Comparing Traditional CCTV with Proposed System

Feature	Traditional CCTV	Proposed System
Video monitoring	Yes	Yes
Automatic person detection	No/limited	Yes
Person counting	Manual/limited	Yes
Object tracking	Limited	Yes
Crowd-density estimation	Limited	Yes
Movement analysis	Mainly manual	Automated
Congestion detection	Manual	Automated
Real-time alerts	Limited	Yes

The proposed system therefore extends conventional surveillance from simple video observation to automated crowd analysis and decision support.

4.7 Strengths & Limitations

Strengths

- Provides automated real-time crowd monitoring.

- Uses YOLO for efficient person detection.
- Supports object tracking.
- Provides crowd-density information.
- Analyses spatial distribution of people.
- Analyses crowd movement.
- Provides threshold-based alerts.
- Reduces dependence on continuous manual observation.
- Can be adapted to different public environments.
- Can be implemented using commonly available computing resources.

Limitations

- Detection accuracy may decrease in extremely dense crowds.
- Severe occlusion can affect tracking.
- Poor lighting can reduce detection performance.
- Camera position and angle influence results.
- Low-resolution video can reduce detection accuracy.
- Real-time processing may require GPU resources.
- Incorrect threshold selection can result in false alerts.
- The system does not guarantee prediction of actual accidents.
- The prototype does not perform facial recognition or personal identification.

4.8 Project Planning, Roles & Budget

Table No 4.6 Project Timeline

Phase	Activity	Duration
1	Problem identification	Week 1
2	Literature review	Week 2
3	Requirement analysis	Week 3
4	System architecture and design	Week 4
5	Dataset/video collection	Week 5
6	YOLO implementation	Week 6–7
7	Tracking implementation	Week 8
8	Density and movement analysis	Week 9

9	Alert mechanism	Week 10
10	Testing and debugging	Week 11
11	Results and evaluation	Week 12
12	Documentation and presentation	Week 13

Team Roles

For a **4-member team**, you can use:

Student Assigned Responsibility Actual Contribution

Student 1 System design & YOLO Person detection and architecture

Student 2 Tracking Object tracking and movement analysis

Student 3 Density & Alert Crowd-density estimation and alerts

Student 4 Testing & Documentation Dataset, testing, graphs and report

Replace “Student 1–4” with your actual names and contributions.

Budget

Item	Estimated Cost	Actual Cost
Existing Laptop/Computer	₹0	₹0
Python	₹0	₹0
OpenCV	₹0	₹0
YOLO	₹0	₹0
Google Colab / Development Environment	₹0	₹0
Existing Webcam	₹0	₹0
Dataset / Sample Videos	₹0	₹0
Internet / Miscellaneous	₹[XX]	₹[XX]
Total	₹[XX]	₹[XX]

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The project “An Intelligent Real-Time Crowd Dynamics Analysis and Monitoring System using Computer Vision” was developed to address the challenges associated with monitoring large and dynamic crowds in public environments. Traditional crowd monitoring mainly depends on manual observation, which can become difficult when large numbers of people and multiple surveillance feeds need to be monitored continuously.

The proposed system provides an automated approach by combining computer vision, deep learning-based person detection, object tracking, crowd-density estimation, spatial analysis, and movement analysis. YOLO is used to detect people from surveillance video, while tracking is used to observe their movement across consecutive frames. The detected information is further used to identify high-density and potentially congested crowd conditions and generate alerts based on predefined thresholds.

The system was implemented as a software-based prototype and tested using different crowd scenarios. The testing and verification process considered important parameters such as person detection, counting, tracking, crowd-density estimation, movement analysis, alert generation, and real-time processing. The results demonstrate the potential of the proposed approach to provide automated crowd information and improve situational awareness compared with conventional manual monitoring.

Although factors such as occlusion, lighting conditions, camera position, crowd density, and computational resources can affect system performance, the developed prototype provides a practical foundation for intelligent crowd monitoring. Overall, the project demonstrates how computer vision can be applied to support safer and more efficient monitoring of crowded environments.

5.2 Future Work

The proposed system can be further improved in several ways:

1. Advanced Object Tracking

More robust tracking algorithms can be integrated to improve tracking accuracy when people overlap or temporarily disappear.

2. Adaptive Crowd-Density Thresholds

Instead of using fixed thresholds, adaptive thresholds can be developed based on the location, time, crowd type, and historical crowd patterns.

3. Advanced Behaviour Analysis

Future versions can analyse more complex behaviours such as sudden crowd dispersal, panic-like movement, running, or reverse-direction movement.

4. Crowd-Density Prediction

Machine learning models can be used to predict future crowd-density changes and provide earlier warnings.

5. Multiple-Camera Integration

Multiple camera feeds can be combined to monitor larger areas and reduce blind spots.

6. Edge Computing

The system can be deployed on edge devices so that video processing occurs close to the camera, reducing latency and unnecessary data transmission.

7. Improved Performance in Dense Crowds

Specialized crowd-analysis and density-estimation models can be incorporated to improve performance when individuals are heavily occluded.

8. Real-Time Monitoring Dashboard

A web-based dashboard can be developed to display crowd count, density, movement direction, alerts, and historical trends.

9. Privacy-Preserving Processing

Future implementations can incorporate techniques such as automatic face blurring and minimal data retention to strengthen privacy protection.

10. Real-World Deployment

The prototype can be evaluated in real environments such as transportation hubs, stadiums, educational institutions, and large public events.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Learned the fundamentals of Computer Vision-based crowd analysis.
- Learned how YOLO object detection can be used to detect people in video.
- Learned the importance of object tracking for analysing movement across video frames.
- Learned methods for estimating and interpreting crowd density.
- Learned how to design threshold-based real-time alert mechanisms.
- Learned about the effects of occlusion, lighting, camera angle, and video quality on computer vision systems.
- Gained understanding of privacy, safety, security, and ethical considerations in AI-based surveillance systems.

Engineering & Professional Skills Developed

- Problem identification and engineering analysis.
- Computer vision and Python programming.
- System architecture and modular design.
- Dataset and video preparation.
- Software testing and debugging.
- Performance evaluation and data analysis.
- Technical documentation and report writing.
- Teamwork and task allocation.
- Engineering decision-making and trade-off analysis.
- Technical presentation and communication.

Individual Reflection

Student 1

The most difficult engineering problem I encountered was implementing reliable person detection under different crowd conditions. I addressed this by studying YOLO-based detection and testing the system with different video inputs. A key engineering decision was selecting YOLO because real-time person detection was an important requirement. I independently improved my understanding of object detection and computer vision through practical implementation. If I repeated the project, I would further improve detection performance for highly crowded and partially occluded scenes.

The most difficult problem I encountered was maintaining consistent tracking of people across consecutive video frames. I solved this by understanding how object tracking associates detections between frames and by testing the tracking module under different movement conditions. A key engineering decision was integrating tracking with person detection because movement analysis requires information over time. I independently learned more about object trajectories and tracking techniques. If the project were repeated, I would investigate more advanced tracking algorithms to improve performance during heavy occlusion.

The most difficult engineering problem I encountered was determining how crowd density and congestion should be identified. I addressed this by analysing the number and spatial distribution of detected people and applying predefined thresholds. The decision to use density-based threshold analysis provided a simple and understandable method for generating alerts. I independently learned how quantitative crowd information can be extracted from computer vision outputs. If I repeated the project, I would develop adaptive thresholds that change according to the environment.

The most difficult problem I encountered was testing and evaluating the complete system under different crowd conditions. I solved this by preparing different test scenarios and analysing detection, tracking, density, and alert outputs. A key decision was to evaluate the system using measurable performance parameters rather than relying only on visual observation. I independently developed skills in result analysis, visualization, and technical documentation. If I repeated the project, I would conduct more extensive real-world testing using multiple cameras and different environmental conditions.

Important: If your team has different actual responsibilities, change these reflections so they honestly describe what each student personally did.

REFERENCES

For your final report, use IEEE referencing style consistently. Your reference list should include the actual sources you used for the literature review, YOLO, OpenCV, datasets, standards, and any software/resources.

Suggested IEEE References

[1] YOLO / Object Detection

G. Jocher, A. Chaurasia, and J. Qiu, *Ultralytics YOLO*, Ultralytics, open-source software.

[2] OpenCV

G. Bradski, “The OpenCV Library,” *Dr. Dobb’s Journal of Software Tools*, 2000.

[3] Computer Vision / Deep Learning

R. Szeliski, *Computer Vision: Algorithms and Applications*, 2nd ed. Cham, Switzerland: Springer, 2022.

[4] Object Detection

J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, “You Only Look Once: Unified, Real-Time Object Detection,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 779–788.

[5] Deep Learning

I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016.

[6] Software Quality

ISO/IEC 25010:2023, *Systems and Software Engineering—Systems and Software Quality Requirements and Evaluation (SQuaRE)—Product Quality Model*, International Organization for Standardization, 2023.

[7] AI Risk Management

ISO/IEC 23894:2023, *Information Technology—Artificial Intelligence—Guidance on Risk Management*, International Organization for Standardization, 2023.

[8] Information Security

ISO/IEC 27001:2022, *Information Security, Cybersecurity and Privacy Protection—Information Security Management Systems—Requirements*, International Organization for Standardization, 2022.

[9] Risk Management

ISO 31000:2018, *Risk Management—Guidelines*, International Organization for Standardization, 2018.

Software / Dataset / Online Resources

For every dataset, software library, website, or open-source repository **you actually used**, add a separate IEEE reference. For example:

[10] OpenCV documentation, “OpenCV Documentation,” online documentation.

[11] Ultralytics, “Ultralytics Documentation,” online documentation.

[12] [Your actual crowd dataset name], [dataset platform/creator], [year].

AI-Assisted Resource Acknowledgement

If you used AI tools such as ChatGPT to help with wording, explanation, coding assistance, or report organization, acknowledge this according to your institution's rules. For example:

OpenAI, *ChatGPT*, AI-assisted tool used for academic writing support, explanation of technical concepts, and code-development assistance,

APPENDICES

Appendix A – Detailed Calculations

This appendix contains the important calculations used in the proposed crowd-monitoring system.

A.1 Crowd Density Calculation

Crowd density can be calculated using:

$$D = \frac{N}{A}$$

where:

- DD = Crowd density in people/m²
- NN = Number of detected people
- AA = Monitored area in m²

Example:

If 40 people are detected in an area of 20 m²:

$$D = \frac{40}{20} = 2 \text{ people/m}^2$$

A.2 Detection Accuracy

If required, detection accuracy can be calculated as:

$$\text{Accuracy} = \frac{\text{Correct Detections}}{\text{Total Actual People}} \times 100$$

Example:

If 90 people are correctly detected out of 100 actual people:

$$\text{Accuracy} = \frac{90}{100} \times 100 = 90\%$$

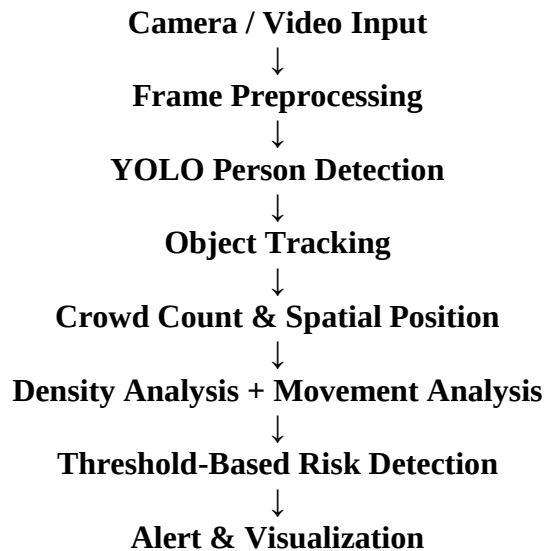
Put your actual calculations and values here. Don't use the example values as your project results.

Appendix B – Engineering Drawings / Schematics

This appendix contains the technical diagrams used in the project.

B.1 System Architecture

Recommended architecture:



B.2 System Flowchart

B.3 User Interface / Output Design

Appendix C – Source Code / Code Repository Information

This appendix contains the essential source-code excerpts used to implement the system.

The complete source code may be maintained in the approved project repository, while only the important sections are included in the report.

C.1 Person Detection Code

[Insert essential YOLO detection code here]

C.2 Tracking Code

[Insert essential tracking code here]

C.3 Crowd Density Calculation

[Insert essential density-calculation code here]

C.4 Alert Generation

[Insert essential alert-generation code here]

C.5 Repository Information

Repository: [Insert GitHub/GitLab repository if available]

Repository Link: [Insert actual link]

If you don't have a repository, you can write: “The complete source code is maintained locally as part of the project implementation.”

Appendix D – Datasheets

Since this project is primarily software-based, this appendix can contain the specifications of the hardware used for implementation.

Table No D.1 Computer/Laptop Specifications

Parameter	Specification
Processor	[Your processor]
RAM	[Your RAM]
GPU	[If available]
Operating System	[Your OS]
Storage	[Your storage]

Table No D.2 Camera/Webcam Specifications

Parameter	Specification
Camera Type	[Webcam/CCTV/etc.]
Resolution	[e.g., 1920 × 1080]
Frame Rate	[XX FPS]
Connection	[USB/IP/etc.]

Attach the actual datasheet/specification page if you used a specific camera or hardware device.

Appendix E – Experimental / Raw Data

This appendix contains the raw data collected during testing.

Table No E.1 Crowd Detection Data

Test No.	Actual People	Detected People	Error	Density	Alert
1	[XX]	[XX]	[XX]	Low	No
2	[XX]	[XX]	[XX]	Medium	No
3	[XX]	[XX]	[XX]	High	Yes
4	[XX]	[XX]	[XX]	Critical	Yes

Table No E.2 Processing Performance

Test No.	Video Resolution	FPS	Detection Accuracy	Tracking Result
1	[Resolution]	[XX]	[XX%]	[Result]
2	[Resolution]	[XX]	[XX%]	[Result]
3	[Resolution]	[XX]	[XX%]	[Result]

Recommended:

- Low-density crowd
- Medium-density crowd
- High-density crowd
- Congested crowd
- Alert condition

Appendix F – Ethics / Institutional Approval

This appendix is applicable only if your institution requires formal ethical approval.

For this project, the system is designed for crowd-level analysis and does not require personal identification.

If no formal approval was required, write:

“No separate institutional ethics approval was required for this academic prototype. The project does not perform facial recognition or personal identification. Video data used for testing were limited to project-related crowd-analysis purposes.”

If your college provided an approval letter, attach it here.

Appendix G – Project Meeting / Progress Records

This appendix contains evidence of project development and team meetings.

Table No G.1 Progress Record

Date	Activity Discussed	Outcome
[Date]	Problem identification	Project topic finalized
[Date]	Literature review	Existing methods studied
[Date]	System design	Architecture finalized
[Date]	Implementation	YOLO detection developed
[Date]	Testing	Initial testing completed
[Date]	Results	Performance evaluated
[Date]	Final review	Project documentation completed

G.2 Meeting Evidence

[Insert photographs/screenshots of project meetings, supervisor discussions, or progress reviews if required.]

Appendix H – User / Industry Feedback

This appendix is **optional and applicable only if you obtained actual feedback** from users, faculty, industry personnel, security personnel, or other stakeholders.

H.1 Feedback Summary

Feedback Source	Feedback	Action Taken
[Faculty/User]	[Actual feedback]	[Action]
[Industry/User]	[Actual feedback]	[Action]

[Attach feedback forms/questionnaires if available.]

Don't create fake feedback just to fill this appendix.

Appendix I – Publications / Patents / Competitions / Product Outcomes

This appendix is applicable only if your project resulted in a publication, patent, competition participation, or product outcome.

I.1 Publication

Title: [If applicable]

Authors: [Names]

Conference/Journal: [Name]

Status: [Submitted/Accepted/Published]

I.2 Competition

Competition Name: [Name]

Event: [Name]

Achievement: [If applicable]

I.3 Product Outcome

If your project has a working prototype:

Prototype: Intelligent Real-Time Crowd Dynamics Analysis and Monitoring System using Computer Vision.

If none of these apply, simply write:

Not applicable.

Appendix J – Individual Contribution Evidence

⚠ This is mandatory for your team project, according to your template.

Each student should provide evidence of their individual contribution.

Table No J.1 Contribution Table

Student	Contribution	Evidence
Student 1	System architecture and YOLO implementation	Code/screenshots
Student 2	Object tracking and movement analysis	Code/screenshots
Student 3	Crowd-density and alert module	Code/results
Student 4	Testing, evaluation and documentation	Test data/graphs/report

J.2 Individual Evidence

For each student, include:

- Screenshots of code personally developed
- Screenshots of implementation
- Testing evidence
- Graphs/results prepared
- Documentation sections contributed
- Meeting/progress evidence, where appropriate

Student 1

Name: Pujitha K

Register No.: 192524149

Contribution:

Worked on implementing the YOLO-based person detection module for identifying people in surveillance video. Developed the tracking component to maintain detected individuals across consecutive frames. Tested the module with different crowd conditions and verified detection and tracking outputs. Worked on improving detection performance under partial occlusion. Integrated the detection and tracking outputs with the next module.

Evidence:



Student 2

Name: Sravani B

Register No.: 192324203

Contribution:

Developed the crowd-density estimation module using the detected number and spatial distribution of people. Implemented analysis of crowd movement and changes in movement patterns. Tested the system with low, medium, and high-density crowd scenarios. Defined suitable thresholds for identifying crowded conditions. Integrated density and movement information with the alert module

Evidence:



Student 3

Name: Hashini S

Register No.: 192524035

Contribution:

Developed the monitoring and alert-generation module for identifying potentially unsafe crowd conditions. Implemented threshold-based alerts for high-density and abnormal movement situations. Designed the output visualization to display detection results and crowd information clearly. Conducted system testing and collected performance results. Evaluated the system based on detection accuracy, processing speed, and alert performance.

Evidence:



Which Appendices Do You Actually Need?

For your **Computer Vision crowd-monitoring project**, I recommend:

Appendix	Need?
A – Detailed Calculations	✓ Yes
B – Drawings/Schematics	✓ Yes
C – Source Code	✓ Yes
D – Datasheets	✓ If hardware used
E – Raw Data	✓ Yes
F – Ethics/Approval	⚠ If applicable
G – Meeting Records	✓ If required by college
H – User/Industry Feedback	⚠ If available
I – Publications/Competitions	⚠ If applicable
J – Individual Contribution	★ Mandatory

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)